# BUILD

## — WITH —

# भारत 2.0

NATIONAL LEVEL HACKATHON

**TEAM NAME:** Team Paradox

**PROBLEM STATEMENT:** Smart Digital FIR & Complaint Management System

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# PROBLEM STATEMENT

## Core Requirement

Develop a secure, accessible and intelligent digital system that simplifies complaint/FIR registration for citizens, while giving police authorities centralized tools for case management, tracking, evidence handling and AI-powered information retrieval from digitized records.

## The Problem

Citizens often face difficulties while reporting incidents and interacting with police systems because of:

- Manual and time-consuming complaint registration
- Language barriers between citizens and authorities
- Difficulty in writing detailed complaints
- Lack of convenient voice-based complaint registration
- Limited transparency in complaint / FIR status
- Difficulty managing digital evidence securely
- Delayed identification of emergency situations
- Fragmented police station-level case management

# SOLUTION

We propose a centralized Digital FIR & Complaint Management System connecting citizens, an intelligent AI processing layer, police officers and the police station head into one accessible, transparent and secure workflow.

### Citizen

Complaint registration, voice input, language selection, evidence upload and status tracking.

### Intelligent Processing Layer

Speech-to-text, translation, classification and emergency detection.

### Police Officer

Verification, FIR registration, investigation, evidence handling and status updates.

### Police Station Head

Centralized database, officer assignment, monitoring and analytics.

# FLOW OF SOLUTION

## End-to-End System Flow

Citizen → AI Processing → Police Officer → Station Head → Case Resolution

## Emergency Detection Pipeline

Voice / Text → Speech-to-Text → NLP Analysis → Keyword + Context → Priority Classification → Dashboard Alert

## Example

**Normal** *"My bike was stolen yesterday."*

**Critical** *"Someone is attacking me right now. I need help."*

*Voice & language intelligence: Citizen speaks (e.g. Marathi) → Speech-to-Text → Translation → structured complaint with category, object, location, language & priority — original audio is always preserved.*

# TECH STACK

### Frontend

- Android / Web Application
- HTML, CSS, JavaScript / React

### Backend

- Java Spring Boot
- or Python FastAPI

### Database

- MySQL
- PostgreSQL

### AI / ML

- Speech-to-Text & Language Detection
- Machine Translation
- NLP Classification & Priority Detection

### Storage

- Secure cloud / object storage for evidence

### Security

- HTTPS, Authentication, RBAC
- Encryption & Password Hashing
- Audit Logging, File Integrity Hashing

# UNIQUE SELLING PROPOSITION (USP)

**01**

### Complaint Registration

Digital submission of complaints with full incident details.

**02**

### Status Tracking

Track complaint / FIR progress live using a unique ID.

**03**

### Multilingual Support

Multiple languages to reduce communication barriers.

**04**

### Voice Complaint

Record in audio; MP3 → Speech → Text conversion.

**05**

### Emergency Detection

Flags potentially urgent complaints for police attention.

**06**

### Secure Evidence Storage

Photos, videos, audio & documents securely tied to cases.

**07**

### Document Digitization

Scans physical police documents into searchable records.

**08**

### AI Police Assistant

Natural-language Q&A over the authorized case database.

**09**

### Centralized Dashboard

Monitor cases, officers and station-level statistics.

# FEASIBILITY & COMPETITORS

## Feasibility

- Built on widely available, production-proven technology (Spring Boot / FastAPI, MySQL / PostgreSQL, standard cloud storage).
- Designed to integrate with authorized government / police systems rather than replace them.
- Core AI components (speech-to-text, translation, NLP classification) are established, off-the-shelf capabilities.
- Role-based access (Citizen, Officer, Station Head) maps directly onto existing police station workflows.

## Key Challenges

- Multilingual speech recognition accuracy
- Emergency detection accuracy
- Evidence security
- False emergency alerts
- Legal & procedural compliance
- Scalability

# RESEARCH & REFERENCE

## Core Database Entities

User | Citizen | Police Officer | Police Station | Complaint | FIR | Investigation | Evidence | Status History | Notification | Audit Log

*A full backend data model — not just a UI.*

## Goals

**Accessibility —** Make reporting easier for every citizen.

**Transparency —** Let citizens & authorized personnel track case progress.

**Security —** Protect sensitive complaints and digital evidence.

**GitHub Repository:** Click here to view our GitHub Repository

**Demo Link:** Demo link